Design and Simulation of a Battery Electric Vehicle Model for Underground Mining Applications with Synthetic Data Generation for Machine Learning Model

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***Abstract*— The shift to electrifying underground mining operations poses considerable challenges in accurately predicting energy consumption due to the intricate nature of operations and the scarcity of real-world performance data for mining electric trucks. This paper introduces a comprehensive dataset synthesis framework as well as the application of four machine learning techniques aimed at forecasting energy usage in electric mining trucks used underground. This addresses the essential need for data-driven optimisation tools in sustainable mining practices. Four machine learning models, including decision tree, random forest, extreme gradient boosting, and deep neural network, were thoroughly assessed using the synthetic dataset comprising 50,000 operational scenarios created through physics-based modelling involving 11 operational parameters. Extensive hyperparameter tuning was conducted using 5-fold cross-validation with a focus on minimising mean absolute error. The Deep Neural Network exhibited outstanding predictive performance ($R^2 = 0.9998$, $RMSE = 1.02$ kWh, $MAE = 0.60$ kWh), showcasing strong generalisation across various operational scenarios. Model validation against physics-based calculations for a typical haul profile confirmed prediction accuracy. This framework provides mining operators with reliable tools to manage energy-efficient fleets, optimise routes, and plan operations.**

I. INTRODUCTION

The underground mining sector is undergoing a significant transition towards electrification driven by more stringent environmental regulations, rising energy costs, and the need to improve safety for workers in confined spaces [1]. The shift from diesel-powered vehicles to battery electric vehicles (BEVs) in underground mining marks one of the industry's most significant technological changes, with global mining firms investing billions in electrifying their fleets [2], [3]. Despite growing adoption, accurate prediction and management of the energy consumption of BEVs in underground conditions is not yet adequately explored. Existing methods are largely developed for surface operations or conventional vehicles. Underground applications introduce additional complexities, e.g., variations in gradient, restricted driving profiles, and limited data availability. Addressing these challenges is crucial for improving energy efficiency, extending battery life, and optimising overall mine productivity.

Efforts to model vehicle energy consumption in mining have evolved through three main stages. The first stage focused on diesel equipment, where early studies established empirical relationships between fuel consumption and operational variables such as payload, haul distance, and road gradient [4]. These models provided foundational insights, but were inherently limited to combustion-based systems, and directly applying these diesel-based formulations to electric drivetrains was inadequate, as they did not represent regenerative braking, battery dynamics, and the high efficiency of electric powertrains [5].

The second stage introduced physics-based models for electric trucks based on detailed energy balance equations and component-level efficiencies [6]–[10]. These models enhanced physical interpretability and captured drivetrain and battery characteristics. However, their application to underground mining has proven challenging. They require precise knowledge of numerous parameters that fluctuate during operations and are difficult to measure in underground mining [11]. Even with a detailed physics-based model, it can be computationally expensive to run model simulations, limiting their practicality for real-time applications [12]. Moreover, as underground mines increasingly generate large volumes of operational data [13], it becomes difficult to continuously update these models to maintain predictive accuracy. Consequently, physics-based models alone struggle to capture the nonlinear and dynamic behaviour of the energy consumption of mining trucks. Regenerative braking adds complexity, allowing electric trucks to recover energy during the downhill or deceleration phases. This introduces nonlinear feedback between velocity, gradient, and battery state of charge, further complicating purely physics-based formulations and motivating hybrid or data-driven approaches.

The third stage of development has seen a growing adoption of machine learning (ML) methods to model vehicle energy consumption [14]–[16]. ML algorithms such as Decision Trees (DT), Random Forests (RF), Extreme Gradient Boosting (XGB), and Deep Neural Networks (DNN) have demonstrated a strong capability to learn complex nonlinear relationships from operational data. In [17], [18] showed that RF and XGB achieve superior predictive accuracy in electric

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vehicle applications. Neural networks have also shown potential to model nonlinear and interactive patterns in vehicle dynamics [19]–[21]. Despite these advances, applications of ML for underground electric haul trucks energy prediction remain limited and fragmented, primarily due to the lack of high-quality, domain-specific datasets.

Access to real-world operational data in underground mining remains a major obstacle to developing accurate energy models. This limitation arises from proprietary restrictions, the novelty of electrification, and the harshness of the operating environment, which hinders consistent data logging and sharing. Flores-Castañeda et al. [22] highlight that although digital technologies are being rapidly adopted in mining, data-sharing practices remain constrained, slowing research progress and innovation.

To mitigate this data scarcity, synthetic data generation has emerged as a promising alternative. Synthetic datasets can simulate realistic mining scenarios without compromising proprietary or sensitive information. Nikolenko et al. [23] emphasise that synthetic data is being increasingly used in domains such as autonomous vehicles and industrial systems, to overcome challenges in data privacy and scarcity. In the mining context, Gürsakal et al. [24] demonstrated that integrating physics-based modelling and statistical sampling techniques can produce synthetic datasets that enhance decision-making and model training. However, validating synthetic datasets against real operational data remains an open research challenge, and few studies have systematically evaluated their effectiveness in electric vehicles.

This research addresses two key challenges: the limited availability of operational data and the need for reliable data-driven modelling frameworks for underground battery-electric haul trucks. While machine learning techniques have been widely applied to energy prediction in light-duty and surface electric vehicles, their applicability to heavy-duty underground mining trucks operating under steep bidirectional gradients, high payload variability, and nonlinear traction remains largely unexplored. To bridge this gap, four ML algorithms - Decision Tree (DT), Random Forest (RF), Extreme Gradient Boosting (XGB), and Deep Neural Networks (DNN) are systematically evaluated for segment-level energy prediction. A physics-consistent synthetic dataset is generated to realistically represent underground haul cycles, compensating for the scarcity of real-world data while preserving operational variability. The study further investigates the influence of key operational parameters on energy consumption and evaluates model generalisability across diverse haul profiles

II. Physics-based model

The electrification of underground mining trucks is a recent trend in the industry, and as such the actual operational data is either not available or not disclosed by the manufacturers. This is due to the technology being in the initial assessment phase and the potential risk of harming their reputation. As a result, physics-based simulation often provides the only practical way to generate training data for ML models. In this work, a modular model is built to generalise well for ML purposes, as shown in Fig 1. The model integrates vehicle, powertrain and battery efficiencies to calculate forces, torque, and power demand under varying operational conditions. By simulating realistic haul profiles and gradients, the model produces synthetic datasets representing energy consumption and generation.

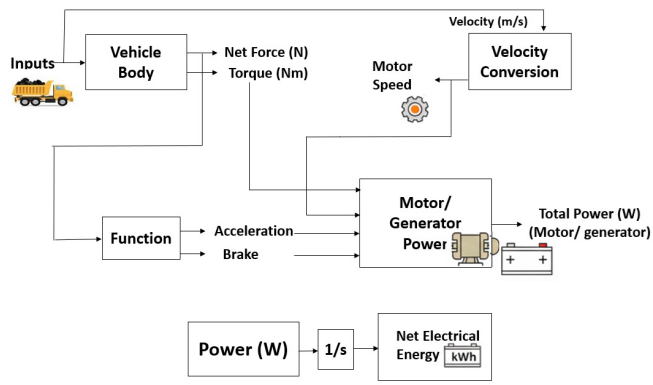


Fig. 1: **Physics-Based Model.** Physics-based model used to generate synthetic data for training and evaluating the machine learning models.

III. Data Preparation

A. Dataset Generation and Preprocessing

The dataset used in this study comprises 50,000 synthetic operational samples, which were generated using a physics-based simulation model designed to replicate the energy consumption and generation patterns of underground electric mining trucks operating in underground environments in typical haul profiles (*flat, uphill, downhill*). Each sample represents a unique operational scenario defined by eleven input features. The upper and lower bounds of these inputs were selected based on manufacturer specifications and published benchmarks [25]–[27], ensuring realistic coverage of vehicle operating conditions. For data generation, random values were sampled from a uniform distribution within these ranges to represent a wide variety of possible operating states without bias toward any specific condition, as shown in Fig. 2.

1) Vehicle mass including payload (m): total gross weight in kg.
2) Rolling resistance coefficient (C_{rr}): terrain-dependent resistance.
3) Aerodynamic drag coefficient (C_d): air resistance factor of the truck.
4) Frontal area (A): cross-sectional frontal surface of the truck affecting drag in (m^2).
5) Terrain slope (%): gradient of the mining route.
6) Wheel radius (r_{wheel}): influences rolling dynamics in (m).
7) Reduction ratio (R_r): ratio of the rotational speed of the motor to the rotational speed of the wheels.

8) Drivetrain efficiency ($\eta_{\text{drivetrain (\%)}}$): mechanical transmission efficiency.
9) Vehicle speed (v): instantaneous haul velocity (m/s).
10) Distance travelled (d): length of the route segment (m).
11) Regeneration battery efficiency ($\eta_{\text{battery (\%)}}$): energy conversion efficiency during regeneration.

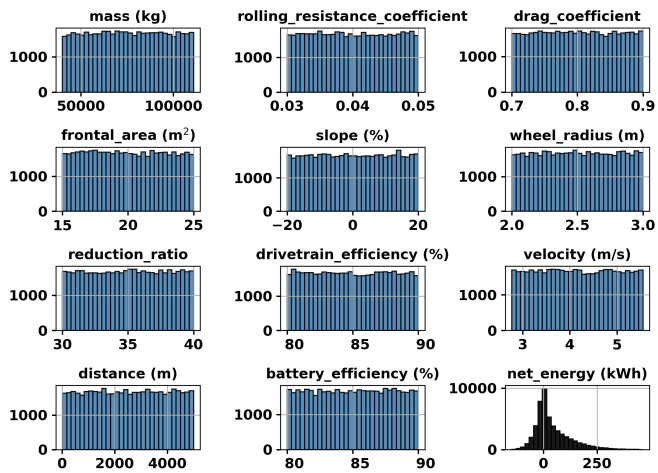


Fig. 2: **Feature Distributions in Synthetic Dataset.** The 11 chosen input features demonstrate realistic operational variability aligned with underground mining conditions, and the output net energy is also shown.

To improve the accuracy of the energy flow model, the electric motor/generator efficiency map (Fig. 3) is incorporated to determine the motoring and generating efficiency ($\eta_{\text{motoring (\%)}}$), ($\eta_{\text{generating (\%)}}$) between different operating points. In this case, it is assumed that the motoring and generating efficiencies are equal. In addition, the gravitational constant (g) is 9.81 m/s^2, and the air density (ρ) is 1.225 kg/m^3. The target variable is the net energy consumption in kWh at the battery, derived from forces, power flows, and drivetrain efficiencies.

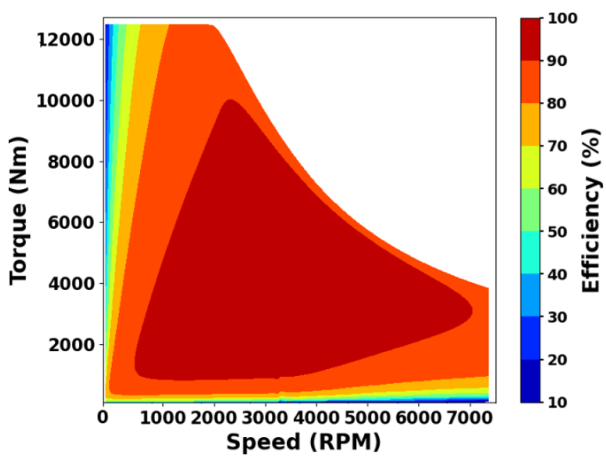


Fig. 3: **Speed-Torque Efficiency Curve.** The efficiency profile illustrates operational performance across varying motor/generating speeds and torque.

Before model training, a Pearson correlation analysis (Fig. 4) was conducted to understand the relationships between the input variables and energy consumption. This analysis highlights terrain slope, vehicle mass, and travel distance as the most influential operational parameters. Following this, the dataset underwent preprocessing for modelling. All input features and outputs are normalised to have a mean of zero and a variance of one to stabilise the training process. After training, the predictions were denormalised to ensure interpretability in real-world applications.

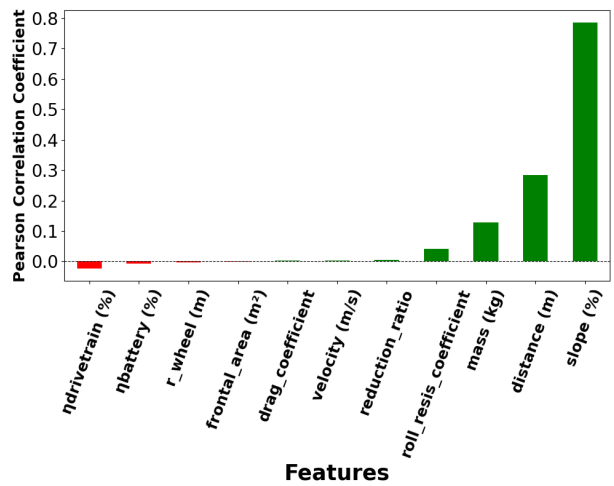


Fig. 4: **Feature Correlation Analysis.** Green colors indicate positive correlations; red colors indicate negative associations

B. Machine Learning Modelling

To assess predictive performance, we evaluated four supervised regression models:

- Decision Tree — A DT is a machine learning model that makes decisions by splitting data into branches based on certain rules such as a flowchart of if–then conditions. It is a simple ML technique to understand and interpret, but can sometimes overfit (learn noise instead of patterns) if the tree becomes too deep [28].
- Random Forest — An RF is an ensemble of many DTs working together. Each tree makes its own prediction, and the forest takes an average vote (for regression) or a majority vote (for classification). This approach reduces the chance of overfitting and improves accuracy compared to a single DT. Although it may not be as interpretable as a decision tree, it still provides insight on the significance of the features, making it interpretable to some measure [29].
- Extreme Gradient Boosting — XGB is an advanced version of a boosting algorithm. It builds trees sequentially, where each new tree tries to correct the errors made by the previous ones. It is designed for high performance, speed, and stability. It often achieves some of the best results for structured/tabular data. However, it is more complex and less interpretable than RF [30].

- Deep Neural Networks — A Neural Network is a supervised learning algorithm used for nonlinear data sets with a large number of inputs. The network architecture comprises one input layer, one output layer, and one or more hidden layers. The input layer receives the raw input data (features). Hidden layers transform these inputs through weighted connections and activation functions, learning complex patterns. The network is trained by computing gradients via backpropagation and updating weights using a gradient-based optimiser. to minimise prediction error. The output layer generates predictions based on the final representation of the last hidden layer [31].

IV. Results and Discussion

A. Model Training and Evaluation

The four ML models (DT, RF, XGB, and DNN) were trained to predict energy consumption in electric haul trucks operating within underground mining environments. All experiments were implemented in `Python 3.11` using the `Scikit-learn` and `XGBoost and Tensorflow` libraries. The dataset was divided into 80% for training and 20% for testing.

Hyperparameter tuning was performed using an exhaustive grid search coupled with k-fold cross-validation ($k = 5$) to minimise error and prevent overfitting. The results are presented in Table I.

TABLE I: Optimised Hyperparameters for Each Model

Model	Optimised Configuration
DT	Max depth = 7; Min samples split = 5; Min samples leaf = 2
RF	Number of estimators = 200; Max depth = 15
XGB	Learning rate = 0.1; Max depth = 6; Number of estimators = 200
DNN	Input: 11 normalised features; Hidden layers: [64, 128, 64] with ReLU activations; Output: 1 neuron; Optimiser: Adam ($\eta = 10^{-4}$); Epochs: 1000

The model accuracy was assessed using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and the Coefficient of Determination (R^2).

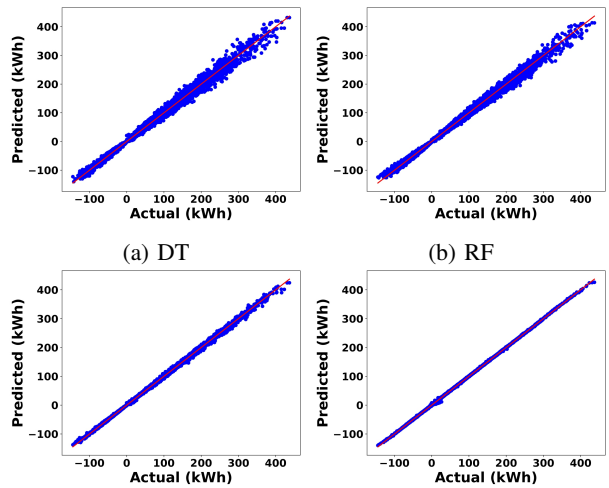
B. Model Performance

The training and testing metrics of the four models are shown in Table II The DNN achieves the best overall performance, with the lowest MAE (0.60 kWh) and RMSE (1.01 kWh) in the test set, while maintaining near-perfect R^2 (> 0.999). The minimal gap between training and testing errors indicates strong generalisation and low overfitting risk, this gap is significantly higher in other models.

Fig. 5 and Fig. 6 present the comparative performance of different machine learning models in predicting energy consumption. As shown in Fig. 5, the predicted values from all models are closely distributed around the around the 1:1 reference line, demonstrating strong predictive agreement with the measured values. Among the evaluated models, the DNN exhibits the smallest dispersion and the highest conformity to the reference line, indicating superior predictive accuracy. The XGB model also achieves high performance, followed by the RF and DT, which show comparatively larger deviations.

TABLE II: **Performance comparison of trained models.** Metrics computed on denormalised predictions (kWh).

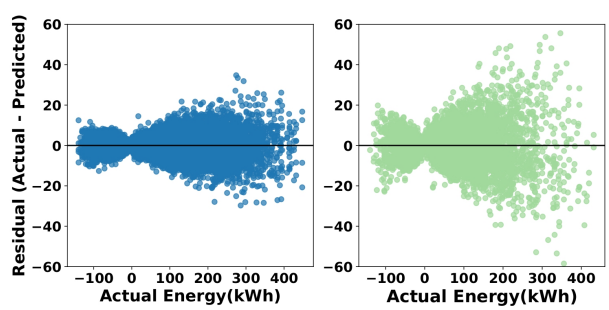
Model	MAE		RMSE		R^2	
	Train	Test	Train	Test	Train	Test
DT	1.84	4.51	3.08	7.32	0.9985	0.9922
RF	2.91	3.67	4.65	5.93	0.9967	0.9949
XGB	**0.48**	1.56	**0.72**	2.56	0.9999	0.9990
DNN	0.52	**0.60**	0.95	**1.01**	**0.9999**	**0.9998**



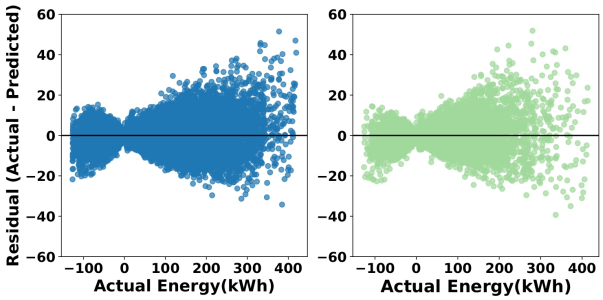
(a) DT (b) RF (c) XGB (d) DNN

Fig. 5: **Actual vs. Predicted Energy Consumption (kWh)** for four models. DNN demonstrates the strongest agreement with ground truth.

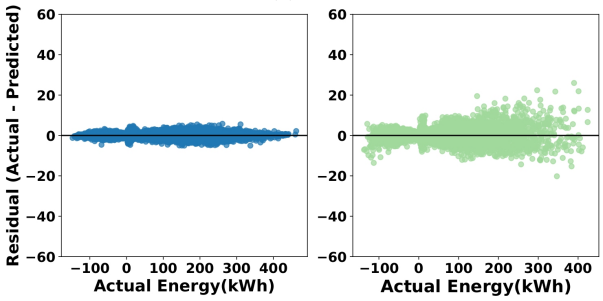
To further assess predictive performance, residual plots Fig. 6 were generated for all approaches, including the DNN and baseline methods such as DT, RF, and XGB. The residual analysis revealed that the DNN consistently outperformed the other models, exhibiting lower error variance and tighter clustering of residuals around zero. These results highlight the DNN's ability to capture the nonlinear relationships between input features and energy consumption more effectively than traditional approaches.



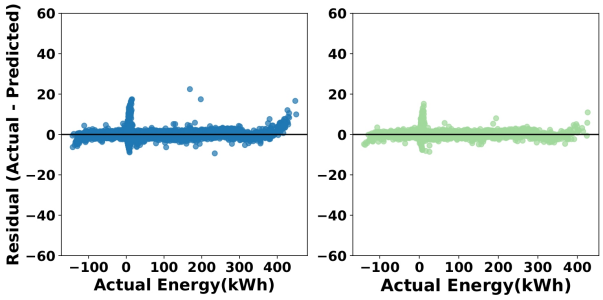
(a) DT



(b) RF



(c) XGB



(d) DNN

Fig. 6: **Residual plots** for four models, Train residuals (blue) and Test residuals (green). DNN outperforms other models.

C. Scenario Analysis

In order to assess both the performance and generalisability of the DNN model, a simulation experiment was carried out using a hypothetical haul profile representing typical underground mining. The haul route consisted of six segments—three for the loaded trip (flat, uphill, and flat) and three for the returning empty trip (flat, downhill, and flat) as shown in Fig. 7 to capture variations in truck loading conditions, terrain gradients, and energy flow mechanisms such as traction demand and regenerative braking.

For each segment, the energy consumption was calculated based on vehicle dynamics, incorporating parameters such as vehicle mass, gradient angle, and rolling resistance.

The results for the haul cycle scenario, where the elevation varied from 0 to 500 m in six steps and other input parameters were assigned specific representative values, are presented in Fig. 8. In all scenarios, the DNN predictions align strongly with the physics-based model (correlation $r > 0.995$), demonstrating high accuracy and robustness.

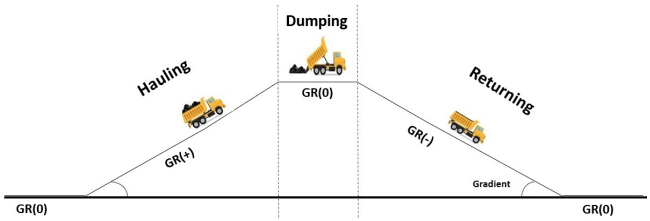


Fig. 7: **Underground mining haul cycles scenario.** Uphill hauling with loaded trucks (positive gradient) and downhill returning with empty trucks (negative gradient).

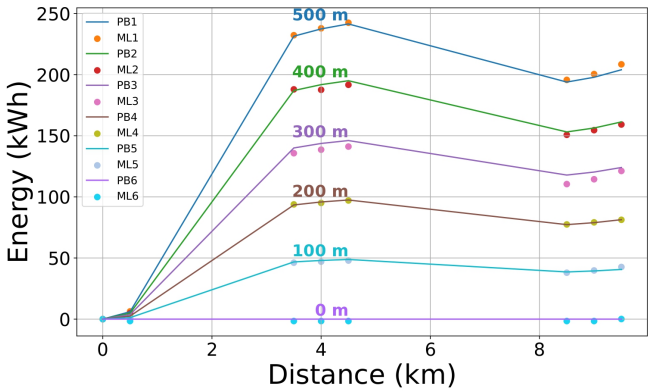


Fig. 8: **Sensitivity Analysis.** Comparison between physics-based calculations (solid lines) and DNN predictions (symbols).

V. Conclusions

This study investigated four supervised machine learning models, namely DT, RF, XGB, and DNN, to forecast the energy consumption of underground mining electric trucks using a physics-based synthetic dataset that replicates actual underground mining conditions. Among these models, DNN demonstrated superior predictive performance, achieving the highest determination coefficient ($R^2 = 0.9998$) and the lowest error metrics (RMSE = 1.01 kWh, MAE = 0.60 kWh). However, the extremely high R^2 values reflect the controlled and physics-consistent nature of the synthetic dataset. In real underground operations, additional sources of uncertainty—such as sensor noise, unmodelled dynamics, and operational variability are expected to reduce predictive performance.

The proposed framework provides a robust foundation for the energy management of underground electric trucks and can be adapted to a wide range of underground mining scenarios. Future research should focus on real-world validation using operational mining data, integrating environmental factors such as temperature, and developing time-series models. The “black box” nature of DNN poses challenges for acceptance in mining, where decision transparency is crucial. Hence, future work will also focus on enhancing the interpretability of DNN models by using explainability techniques like SHAP and uncertainty-aware models like Monte-Carlo dropout to enhance model explainability and trustworthiness.

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